# MCA SEMESTER 2

## Data Structures Question Bank

1. Define: Data
2. Define: Data structures
3. What are data types?
4. Draw the classification of Data Structures.
5. Explain primitive data types.
6. What is linear data structure?
7. What is non-linear data structure?
8. Explain :Stack
9. Explain : Queue
10. What is a linked list?
11. Write difference between linear and non-linear data structures
12. Explain operations performed on different data structures
13. What is algorithm efficiency?
14. Explain time complexity with example.
15. Find the time complexity of the following code: print("Hello World")
16. Find the time complexity of the following code:

```
int i, n = 8;
        for (i = 1; i <= n; i++) {
                printf("Hello World !!!\n");
        }
```

17. Find the time complexity of the following code:

```
Sum(a, b) { return a + b }
```

18. Explain space complexity.
19. Consider the following stack of city names: STACK: Ahmedabad, Surat, Baroda ,____,____. Describe the stack as the following operations take place:

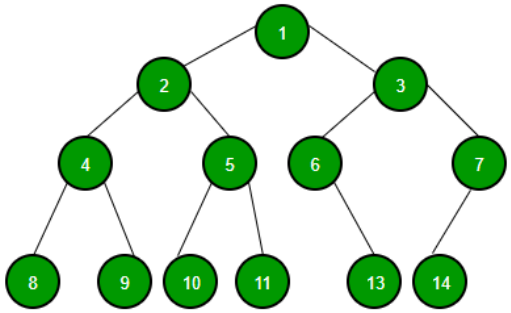
a) PUSH(Mumbai)
b) PUSH(Nagpur)
c) POP()
d) POP()
e) POP()
f) PUSH(Rajasthan)
g) PUSH(Jamnagar)

20. Evaluate the following postfix operation: 5, 3, +, 2, *, 6, 9, 7, -, /, -
21. Evaluate the following postfix operation: 5, 6, 2, +, *, 12, 4, /, -
22. Evaluate the following postfix operation: *2 3 1 * + 9 -*
23. Convert the following expression from infix to postfix. (A + B) / (( D - E) * F )
24. Convert the following expression from infix to postfix. (A + B) * (C + D)
25. Convert the following expression from infix to postfix. A + B + C + D
26. Convert the following expression from infix to postfix. (A + B) * C
27. What is Polish Notation?
28. Define: Stack
29. Define: Queue
30. Draw and explain the structure of singly linked
31. Draw and explain the structure of circluar linked
32. Consider the following simple queue of characters: STACK: A, B, __, __, __, __. Describe the queue as the following operations take place:
    a) Insert(D)
    b) Insert(E)
    c) Insert(F)
    d) Insert(G)
    e) Insert(H)
    f) Delete()
    g) Delete()

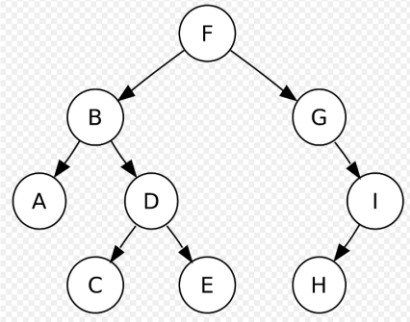
33. Consider the following queue where the queue is circular queue allocated with 6 memory cells. Front = 2, Rear = 5. QUEUE: ___, X, Y, Z, A, ___ Describe the circular queue as the following operations take place:
    A) Insert(B)
    B) Insert(P)
    C) Insert(Q)
    D) Delete()
    E) Delete()
    F) Delete()

34. Write algorithm for evaluation of postfix expression.
35. Write algorithm for stack push operation
36. Write algorithm for stack pop operation
37. Write algorithm for simple queue insert operation
38. Write algorithm for simple queue delete operation
39. Write algorithm for circular queue insert operation
40. Write algorithm for circular queue delete operation
41. Write algorithm to insert an element at the first position in singly linked list
42. Write algorithm to insert an element at the middle position in singly linked list
43. Write algorithm to insert an element at the last position in singly linked list
44. Write algorithm to insert an element at the first position in circular linked list
45. Write algorithm to insert an element at the middle position in circular linked list
46. Write algorithm to insert an element at the last position in circular linked list
47. Write algorithm to insert an element in doubly linked list
48. Write algorithm to delete an element in doubly linked list
49. Write an algorithm to delete the first element in singly linked list
50. Write an algorithm to delete the first element in doubly linked list
51. Write an algorithm to delete the first element in circular linked list

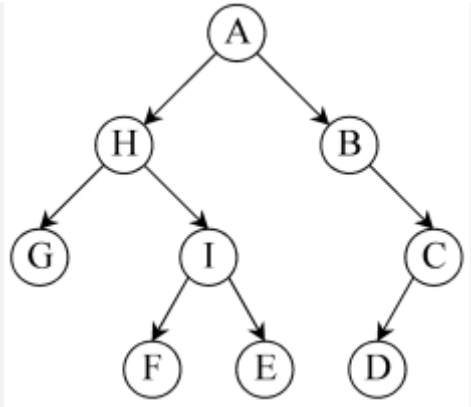
52. Write an algorithm to print all the elements in singly linked list
53. Write an algorithm to print all the elements in doubly linked list
54. Write an algorithm to print all the elements in circular linked list
55. What is a leaf node? Give one example of a tree having leaf node.
56. What is a root node?
57. Define: Edge
58. Identify whether the following tree is a binary tree or binary search tree. (Explain with reason)



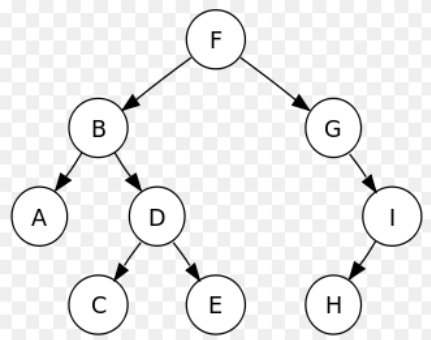

59. Generate a binary search tree from the following: 45,23,65,89,16,28,9,36
60. Find the pre order traversal of the following tree.



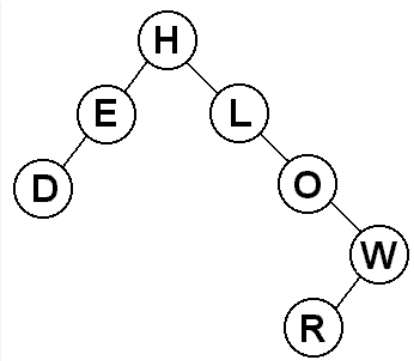
61. Find the In order traversal of the following tree.



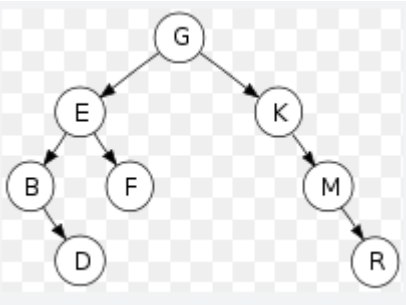

62. Find the Post order traversal of the following tree.



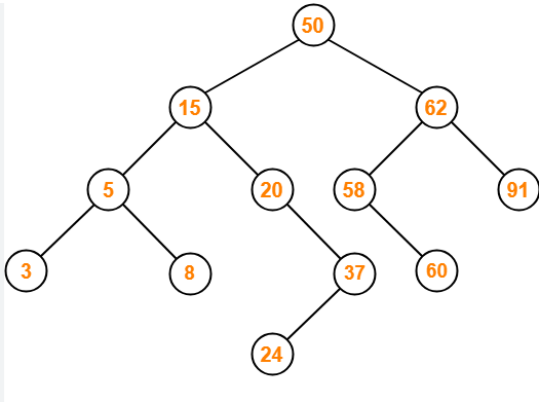

63. Define: Binary search tree. Give one example of it.
64. Create an AVL tree from the following: 14,17,11,6,53,4,13
65. Delete the node W from the following tree.



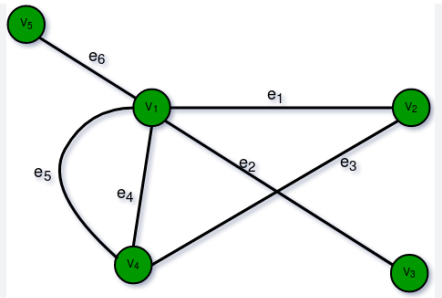

66. Delete the node K from the following tree.



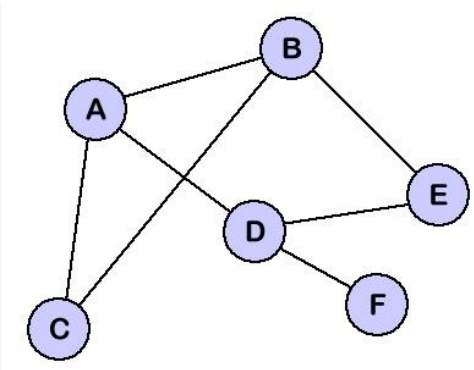

67. Delete the node 50 from the following tree.



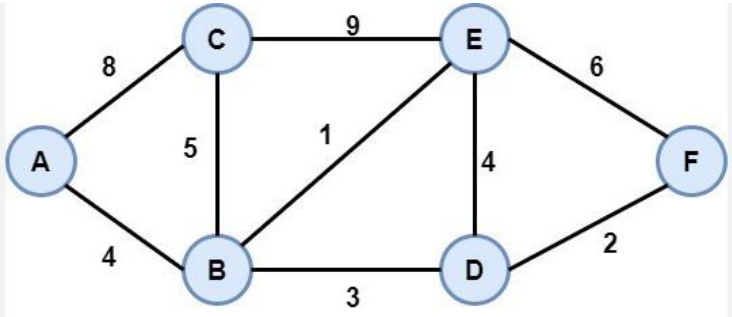

68. What is an AVL tree?
69. What is LL rotation?
70. What is RR rotation?
71. What is RL rotation?
72. What is LR rotation?
73. Explain Btrees.
74. What is a directed and undirected graph. Explain with example.
75. Give one example of closed path.
76. Identify whether the following graph is a directed graph or undirected graph. Find the parallel edges from the graph. Write name of the parallel egde.




77. Write the adjacent nodes of the node A, E and F in the following graph.

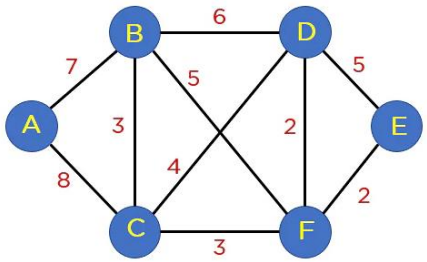



78. Explain closed path and simple path with example.
79. Give an example of complete graph with example.
80. Find the degree of all the E, D, and F from the following graph.

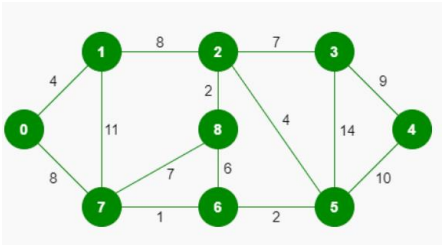
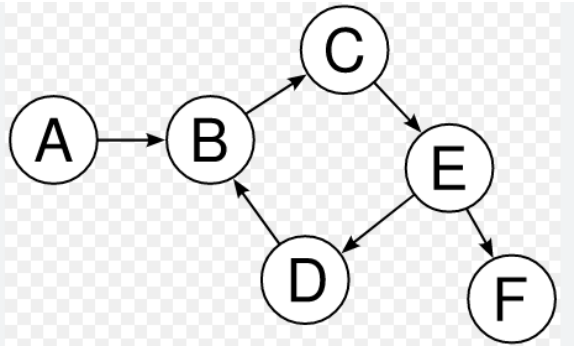



81. Find leaf nodes from the following tree.
82. Generate an AVL tree from the following:64,1,44,26,13,110
83. Give one example of B-tree of order 2.
84. Define: Hashing
85. Explain: Separate chaining
86. Define: Closed hashing
87. Apply bubble sort on the following data and show every pass: 34,21,78,30,71,40,93

88. Apply selection sort on the following data. Show every pass.: 90,23,74,10,5,62,85
89. Apply insertion sort on the following data. Show every pass.: 83,5,10,92,50,81,4
90. Apply merge sort on the following data: 45,86,9,10,52,39,86
91. Write algorithm for bubble sort
92. Write algorithm for insertion sort.
93. Write algorithm for selection sort.
94. Write algorithm for binary search.
95. Apply radix sort and sort the following data: (Show every pass)123,765,920,482,840,721,592,530
96. What are collision resolution techniques?
97. Apply quick sort on the following data: 44,33,11,55,77,90,40,60
98. Apply prim's algorithm and find the minimum spanning tree from the following graph:




99. Apply kruskal's algorithm and find the minimum spanning tree from the following graph:



100. Find the adjacency martix of the following graph: